Reza Ghasemi

Machine Learning specialist

Edmonton ,AB ,Canada (Open to relocating to any province in Canada)
Email: r.ghasemi7171@gmail.com Github: github.com/Reza-CSEE/
UBERG Project: uberg.ca/ VoiceBot Project: github.com/Reza-CSEE/VoiceBot
Voice Detector demo: rezarastin-ai-voice-detector-rastin.hf.space

SUMMARY

Machine Learning specialist with graduate-level expertise in NLP, LLMs, and applied AI. Experienced in designing and optimizing ML models for information retrieval, text analysis, and RAG applications. Proficient in Python, PyTorch, TensorFlow, and data engineering for large structured/unstructured datasets. Collaborative, research-driven, and eager to deliver real-world impact. Also built a signal-level detector for AI-generated voices, validated on seven datasets; open to collaboration and applied audio-AI roles.

SKILLS

NLP | LLMs | C/C++ | Python | PyTorch | TensorFlow | Scikit-learn | Pandas | SQL | Linux | Git | Signal Processing | Audio ML | Spoofing Detection | Spectral Analysis | Statistical Modeling | Gradio | Hugging Face | JavaScript (Web Audio) | Retrieval | Optimization | Deployment | Collaboration | Research | Communication | ProblemSolving

EXPERIENCE

Independent Research | Detection of AI-Generated Voices (Rastin-Voice)

Jun 2025 – Present

- Designed a **signal-level detector** that tells human speech from AI-generated and cloned voices, including voices from systems it was never tuned on, without training a neural network on known voice generators.
- Validated on **7 datasets spanning 5 voice-synthesis architecture families**, reaching **AUC 0.94–0.99** on realistic test sets (including **In-the-Wild** and **ASVspoof 5**), on noisy real-world audio.
- Audited the pipeline for data leakage and fixed several causes of falsely perfect results (shared data pools, silent file-loading failures, sign errors).
- Built and deployed an interactive demo (**Python, Gradio, Hugging Face Spaces**) with uncompressed 32-bit browser recording, a live spectrogram, and a 10-second live verdict.

Computer Science Lab, Bishop's University | UBERG – Environmental Deep Learning Application

Sep 2023 – May 2025

- Contributed to a research initiative on water ecosystems by developing a buoy-based floating lab for algae monitoring.
- Needed to automate algae identification from microscopic images collected in Canadian lakes.
- Trained and fine-tuned **CNNs** models on real datasets; applied preprocessing, augmentation, and hyperparameter tuning.
- Achieved **>92% classification accuracy**, enabling reliable detection of multiple algae species for environmental monitoring.

Computer Science Lab, Bishop's University | Voice Bot Project

Sep 2022 – Aug 2023

- Developed **NLP** and **LLM-based** Voice Bot with **Python**, applying retrieval to enhance query understanding.

- Implemented serverless **AWS** Lambda pipelines with **RDS**, improving scalability and reducing latency by 40%.
- Optimized ML workflows for speech and document retrieval, raising accuracy and overall system efficiency.
- Collaborated with team to enhance performance, increasing task success by 23% and user satisfaction by 27%.
- Improved **SQL-based** behavior analysis to identify drop-offs, reducing user exit rate by 19% during interactions.

Achievements

- **Best Paper Award**, 21st Electrical Distribution Networks Conference, 2016.

- Developed a neural network model predicting regional power outages using real-world data with **98% accuracy**.

EDUCATION

Master's Degree in Computer Science

Bachelor's Degree in Electrical Engineering(Control)